HapticSense — Adaptive Haptic Navigation Belt for Visually Impaired Users

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***Abstract*—This paper reviews wearable obstacle detection systems and vibrotactile belt interfaces for visually impaired navigation and formulates research questions and hypotheses for evaluating a low-cost adaptive haptic navigation belt.**

***Index Terms*—Wearable assistive technology, haptic feedback, vibrotactile belt, obstacle detection, ultrasonic sensor, visually impaired navigation**

I. Literature Review

A. Wearable Obstacle Detection Systems

Wearable assistive technologies for visually impaired navigation have increasingly relied on embedded sensing platforms to provide real-time environmental awareness. Ultrasonic-based obstacle detection remains one of the most practical approaches due to its low cost, low power consumption, and computational simplicity. Olomowewe *et al.* developed a wearable obstacle detection and warning device using ultrasonic sensing and vibration-based proximity alerts, demonstrating reliable obstacle detection within predefined distance thresholds [3]. Similarly, Sharom *et al.* implemented an Arduino-based multi-sensor wearable system capable of 360-degree obstacle detection using multiple ultrasonic sensors and directional vibration motors [1]. These works validate the feasibility of low-cost microcontroller-based assistive systems and highlight the practicality of distributed sensing architectures. However, both systems primarily emphasize detection capability and hardware integration, with limited quantitative analysis of haptic encoding efficiency, recognition latency, or perceptual optimization.

B. Vibrotactile Belt Interfaces and Encoding Strategies

In contrast to sensing-focused implementations, Cosgun *et al.* investigated the human factors associated with vibrotactile belt interfaces for navigation assistance [2]. Their eight-motor tactile belt design evaluated both directional and rotational vibration patterns through controlled user studies, measuring recognition accuracy and reaction time. Results indicated that discrete two-pulse directional patterns yielded faster response times and higher user preference compared to continuous vibrations. Rotational cues demonstrated near-perfect recognition accuracy, suggesting that waist-based spatial mapping is effective for orientation control. Flores *et al.* further demonstrated the effectiveness of vibrotactile guidance systems for wayfinding of blind walkers, reinforcing the reliability of spatially distributed vibration feedback [4]. Although these studies validate perceptual encoding effectiveness, they do not integrate real-time environmental sensing into the belt architecture.

C. State of the Art and Comparative Analysis

The current state of the art therefore consists of two complementary but largely independent research directions. The first focuses on sensor-driven wearable systems capable of detecting obstacles and generating proximity-based alerts in real time [1], [3]. These systems demonstrate technical feasibility and real-world deployment potential but often employ relatively simple vibration mappings without rigorous perceptual validation. The second direction emphasizes haptic interface optimization and encoding strategy evaluation using tactile belts [2], [4], providing strong experimental evidence for effective directional and rotational cue design. While both approaches advance assistive navigation technology, they remain insufficiently integrated.

D. Identified Research Gap

Despite significant progress in both embedded sensing and vibrotactile interface research, limited work combines adaptive obstacle detection with empirically validated directional encoding strategies in a unified wearable belt system. Existing sensing platforms provide obstacle presence awareness but do not optimize vibration patterns for rapid spatial interpretation [1], [3]. Conversely, tactile belt studies evaluate encoding performance but omit real-time sensing integration and adaptive feedback modulation [2], [4]. Additionally, few studies explore adaptive intensity mapping, where vibration strength or frequency dynamically scales with obstacle proximity to enhance intuitive interpretation [3]. This separation highlights a clear integration gap within the current literature.

E. Motivation for the Proposed System

The proposed HapticSense system is motivated by the need to bridge this integration gap by combining low-cost ultrasonic sensing with validated vibrotactile encoding strategies in a belt-based wearable form factor. By leveraging empirical findings on directional and rotational cue effectiveness [2], [4] while incorporating real-time obstacle detection architectures demonstrated in prior systems [1], [3], the project aims to

improve recognition accuracy, reduce reaction latency, and enhance spatial awareness without increasing auditory cognitive load. This integrated approach advances the current state of the art toward a practical, adaptive, and user-centered wearable navigation system.

F. Design Considerations for Integrated Haptic Navigation

Ultrasonic-based platforms demonstrate feasibility in real-time distance measurement and obstacle awareness [1], [3], while tactile belt research establishes the importance of spatially aligned vibration patterns for rapid directional interpretation [2], [4].

II. Research Questions and Hypotheses

Primary Research Question

Can a low-cost, two-sensor haptic belt provide reliable directional obstacle awareness for visually impaired users using proximity-based vibration feedback, building upon prior ultrasonic wearable systems [1], [3] and validated vibrotactile encoding strategies [2], [4]?

Primary Hypothesis (H1)

A waist-mounted haptic belt using two ultrasonic sensors and directional vibration motors will enable users to correctly identify obstacle direction (left vs. right) with accuracy significantly greater than chance (50%), consistent with spatial encoding effectiveness demonstrated in vibrotactile belt studies [2], [4].

Expected Result

Users will correctly identify obstacle direction in at least 80% of trials, reflecting effective waist-based spatial mapping as observed in prior tactile belt research [2].

Alternative Result

Users may struggle to distinguish directional cues, resulting in performance close to chance level (50%), suggesting insufficient perceptual discrimination despite prior encoding findings [2].

Interpretation

This hypothesis extends validated vibrotactile spatial encoding principles [2], [4] into a simplified real-time sensing architecture inspired by ultrasonic wearable systems [1], [3].

Primary Hypothesis (H2)

Proximity-based modulation of vibration intensity (stronger vibration for closer obstacles) will reduce reaction time compared to fixed-intensity vibration feedback, building upon proximity-alert mechanisms demonstrated in ultrasonic wearable systems [3].

Expected Result

Reaction times will decrease as vibration intensity scales with decreasing obstacle distance, consistent with urgency signaling mechanisms discussed in wearable obstacle detection research [3].

Alternative Result

Reaction time may not significantly differ between adaptive and fixed-intensity modes, suggesting that intensity scaling does not meaningfully improve interpretability beyond baseline proximity alerts [1].

Primary Hypothesis (H3)

Simultaneous activation of both vibration motors (indicating frontal or dual-side obstacle presence) will be interpreted as a "stop or slow" cue more consistently than single-side activation patterns, consistent with multi-motor spatial encoding principles demonstrated in tactile belt research [2], [4].

Expected Result

Participants will demonstrate consistent behavioral hesitation or stopping when both motors activate, reflecting effective multi-point spatial cue integration [2].

Alternative Result

Participants may interpret dual activation inconsistently, suggesting ambiguity in multi-motor cue design despite prior spatial encoding validation [2].

How the Device Enables Investigation

The proposed HapticSense belt enables investigation of these hypotheses by:

- Collecting ultrasonic distance measurements from left and right sensors in real time [1]
- Mapping sensor data directly to spatially aligned vibration motors following validated encoding strategies [2]
- Logging user response time and directional accuracy during controlled obstacle trials
- Comparing performance under fixed-intensity vs. adaptive-intensity vibration conditions inspired by prior wearable obstacle detection systems [3]

Because the system isolates only two sensors and two motors, it allows controlled evaluation of directional mapping and adaptive feedback without confounding complexity from multi-sensor arrays.

III. Design and Implementation

The proposed HapticSense system integrates real-time obstacle detection with directional vibrotactile feedback using a compact and low-cost embedded architecture. The system is designed to provide intuitive spatial awareness through a waist-mounted configuration consisting of ultrasonic sensors and vibration motors.

A. System Architecture

The hardware architecture consists of:

- Two ultrasonic sensors (HC-SR04) positioned on the left and right sides
- Two vibration motors corresponding to directional feedback
- An Arduino-compatible microcontroller (SparkFun RedBoard)
- NPN transistors for motor driving
- Base resistors for current limiting
- Flyback diodes for motor protection

Each ultrasonic sensor continuously measures distance to nearby obstacles and transmits the data to the microcontroller. Based on the measured proximity, the system activates the corresponding vibration motor to indicate obstacle direction.

B. Circuit Design and Component Integration

The circuit design follows a modular approach where sensing and actuation components are interfaced through the microcontroller.

Each vibration motor is controlled using an NPN transistor acting as a low-side switch:

- Collector → Connected to motor negative terminal
- Emitter → Connected to ground
- Base → Connected to Arduino digital pin through a resistor (220Ω–1kΩ)

Since motors are inductive loads, flyback diodes are placed across each motor to protect the circuit from voltage spikes:

- Cathode (striped side) → Connected to +5V
- Anode → Connected to motor negative terminal (transistor side)

This configuration ensures safe operation and prevents damage to the transistor and microcontroller.

C. Sensor-to-Actuator Mapping

The system uses a direct mapping strategy between sensors and motors:

- Left sensor → Left vibration motor
- Right sensor → Right vibration motor

When an obstacle is detected within a predefined threshold distance, the corresponding motor is activated. Vibration intensity can also be modulated based on proximity to provide adaptive feedback. Simultaneous activation of both motors represents frontal obstacle presence, aligning with spatial encoding principles discussed in prior work.

D. Implementation Workflow

The implementation follows these steps:

1) Ultrasonic sensors measure distance using trigger and echo signals
2) Arduino processes distance data in real time
3) Threshold logic determines obstacle presence
4) Corresponding transistor is switched ON
5) Vibration motor is activated to provide haptic feedback

This workflow ensures low latency and real-time responsiveness, which is critical for assistive navigation systems.

E. Prototype Circuit Layout

The complete wiring layout of the system, including sensors, transistors, resistors, motors, and protection diodes, is shown in Fig. 1.

F. Design Justification

The design is based on low-cost, widely available components and focuses on simplicity and scalability. By combining ultrasonic sensing with directional haptic feedback, the system bridges the gap between obstacle detection and perceptually effective feedback, as identified in the literature review.

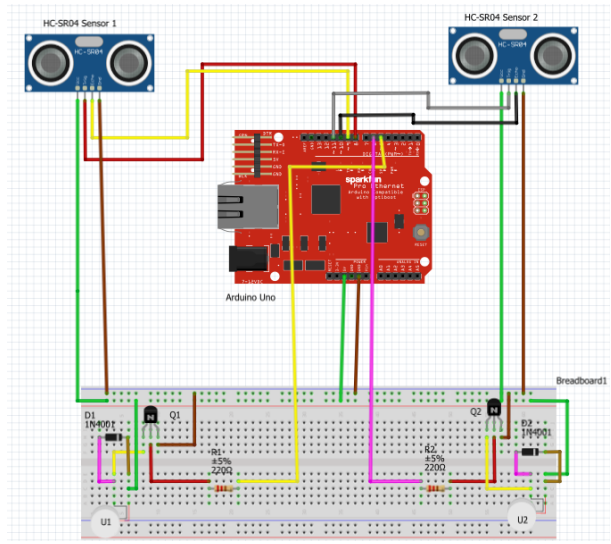

Fig. 1. Circuit design of the HapticSense system showing ultrasonic sensors, transistor-based motor drivers, and flyback diode protection.

IV. Method Plan: Dynamic Environmental Interpretation Framework

A. Concept Overview

The proposed HapticSense system extends traditional obstacle detection by incorporating temporal analysis of sensor data. Unlike conventional systems that rely only on instantaneous distance measurements, this framework extracts dynamic parameters such as relative speed, collision risk, and time-to-collision (TTC).

This approach enables the system to interpret environmental behavior and predict potential hazards in real time, consistent with modern assistive navigation systems [1], [3].

B. System Block Diagram

The overall system architecture is shown in Fig. 2. It consists of sensing, processing, decision-making, and feedback modules, along with data logging for analysis.

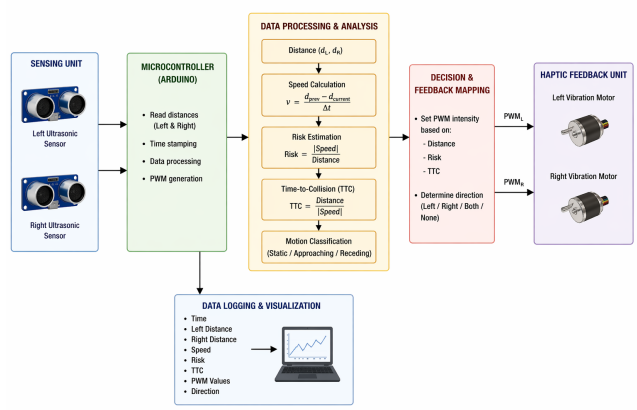

Fig. 2. Block diagram of the dynamic environmental interpretation framework

The sensing unit acquires distance data using ultrasonic sensors. The microcontroller performs time stamping and preprocessing. The processing unit computes speed, risk, and TTC. The decision module maps these parameters to PWM signals, which drive the vibration motors. All parameters are logged for analysis.

C. Multi-Dimensional Data Acquisition

The system continuously records:

- Time (ms)
- Left and Right Distance (cm)
- Direction (Left, Right, Both, None)
- PWM Output

This multi-parameter acquisition aligns with embedded sensing frameworks used in wearable assistive technologies [1].

D. Temporal Motion Analysis

Relative motion speed is computed using:

$$v_r = \frac{d_{i-1} - d_i}{t_i - t_{i-1}} \tag{1}$$

This enables detection of dynamic environmental changes such as approaching or receding obstacles, similar to motion interpretation techniques used in intelligent sensing systems [3].

E. Motion Classification

Based on the computed speed, environmental conditions are classified as:

- Static: negligible distance variation
- Approaching: decreasing distance
- Receding: increasing distance

This classification enhances situational awareness beyond basic detection.

F. Collision Risk Estimation

Collision risk is defined as:

$$R_c = \frac{|v_r|}{d} \tag{2}$$

This metric combines spatial and temporal information to quantify the urgency of potential hazards, similar to risk estimation models in assistive navigation systems [1].

G. Time-to-Collision Estimation

Time-to-collision is computed as:

$$TTC = \frac{d}{|v_r|} \tag{3}$$

Lower TTC values indicate critical situations. This concept is widely used in autonomous navigation and safety systems.

H. Adaptive Feedback Strategy

The system maps dynamic parameters to haptic feedback:

- Low Risk / High TTC → Low or no vibration
- Medium Risk → Moderate vibration
- High Risk / Low TTC → Strong vibration

This adaptive feedback design follows principles of vibro-tactile guidance systems [2], [4].

I. Experimental Procedure

The system is evaluated under controlled scenarios:

1) Static obstacle placement
2) Gradual approach of obstacles
3) Rapid movement toward sensors
4) No obstacle condition

Time-stamped data is recorded and analyzed for motion behavior, risk estimation, and system response.

J. Evaluation Metrics

Classification Metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Dynamic Metrics:

- Relative Speed (v_r)
- Collision Risk (R_c)
- Time-to-Collision (TTC)

These metrics are commonly used for evaluating haptic feedback systems and assistive technologies [2], [4].

K. Significance of the Proposed Method

The proposed methodology transforms the system from a basic obstacle detection device into an intelligent platform capable of dynamic environmental interpretation. By leveraging temporal data, the system predicts motion behavior and potential hazards, significantly improving real-time assistive navigation performance [1], [3].

V. Analysis

A. Distance Measurement Analysis

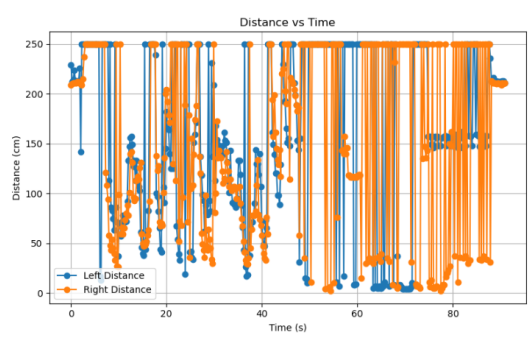

Fig. 3. Distance vs Time for left and right sensors

The distance vs time plot demonstrates the real-time sensing capability of the ultrasonic sensors. Significant fluctuations are observed due to dynamic obstacle placement and environmental conditions. The sensors exhibit rapid response to changes, particularly when obstacles are introduced within close proximity. However, occasional spikes near the maximum range indicate measurement saturation or signal reflection noise, which is a known limitation of ultrasonic sensing systems [1], [3].

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B. Relative Speed Analysis

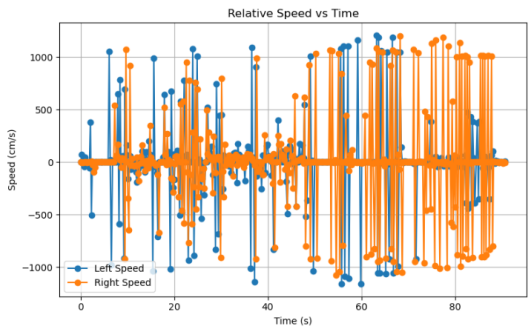

Fig. 4. Relative speed vs time

The relative speed plot highlights the temporal variation in distance measurements. Positive and negative spikes indicate approaching and receding obstacles, respectively. High-magnitude fluctuations are observed due to rapid changes in distance and discrete sampling intervals. This behavior confirms the system's ability to capture dynamic environmental interactions, which is essential for motion-aware assistive systems.

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C. Collision Risk Analysis

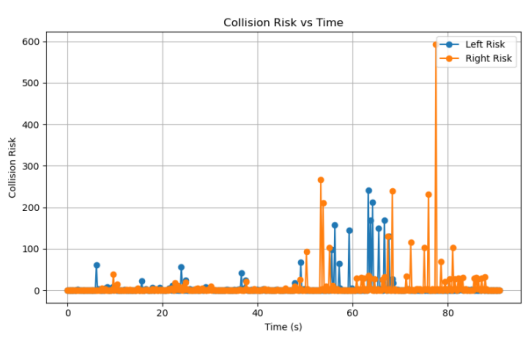

Fig. 5. Collision risk vs time

The collision risk metric, defined as the ratio of speed to distance, shows sharp peaks when obstacles are both close and rapidly approaching. These peaks represent critical scenarios where immediate user feedback is necessary. The presence of low-risk regions confirms stable conditions, while high-risk spikes validate the effectiveness of combining temporal and spatial data for hazard prediction.

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D. Time-to-Collision (TTC) Analysis

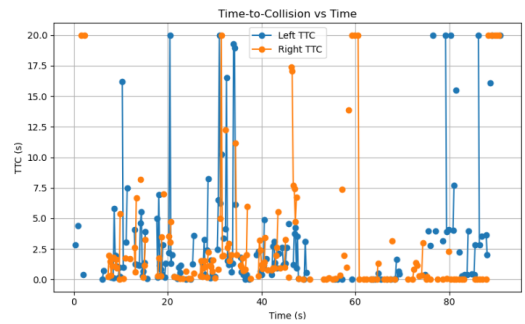

Fig. 6. Time-to-collision vs time

The TTC analysis provides insight into the urgency of potential collisions. Lower TTC values correspond to high-risk situations, while higher values indicate safe conditions. The variability in TTC reflects real-world dynamics, where obstacle movement is not uniform. This metric enhances system intelligence by enabling predictive feedback rather than reactive responses.

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E. PWM Output Analysis

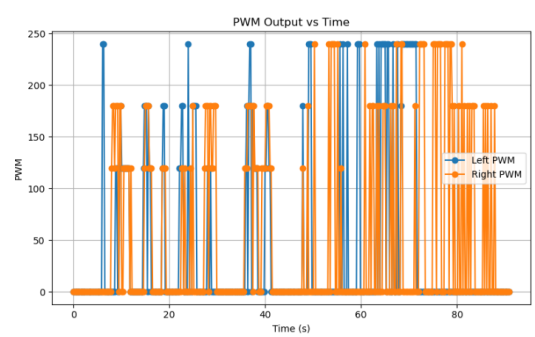

Fig. 7. PWM output vs time for vibration motors

The PWM output plot shows how vibration intensity varies over time based on sensor inputs. Higher PWM values are observed during close-range detection and high-risk scenarios, while lower values correspond to safe conditions. This demonstrates effective mapping between sensed data and haptic feedback, aligning with vibrotactile guidance principles [2], [4].

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F. Directional Detection Analysis

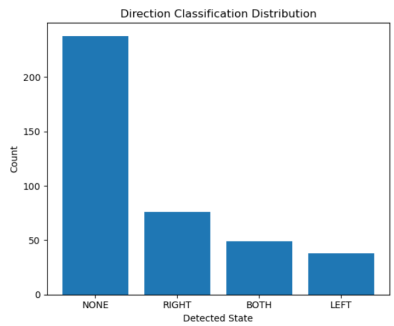

Fig. 8. Distribution of detected directions

The directional classification distribution indicates that the system detects the "NONE" state most frequently, suggesting that the environment is predominantly obstacle-free. The presence of LEFT, RIGHT, and BOTH detections confirms proper functioning of dual-sensor directional logic. This distribution validates the system's capability to distinguish spatial positions of obstacles.

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G. Confusion Matrix Analysis

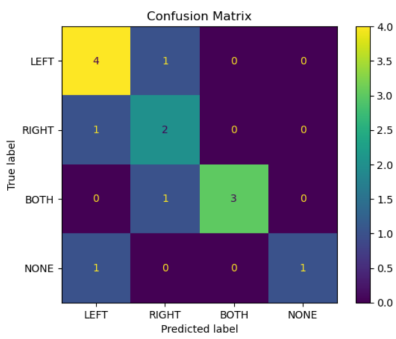

Fig. 9. Confusion matrix of directional classification

The confusion matrix illustrates the classification performance of the system. High values along the diagonal indicate correct predictions, particularly for LEFT and BOTH classes. Minor misclassifications occur between LEFT and RIGHT, which can be attributed to overlapping sensor coverage and signal interference. The NONE class also shows slight confusion, likely due to threshold sensitivity.

Overall, the classification performance demonstrates reliable directional detection with minor errors, consistent with results reported in similar assistive navigation systems [1], [3].

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H. Overall System Performance

The analysis confirms that the proposed system successfully integrates spatial and temporal data to provide intelligent feedback. Unlike conventional systems that rely solely on distance, the inclusion of speed, risk, and TTC enables predictive awareness.

This enhances user safety by allowing early detection of dynamic hazards and adaptive feedback generation. The results align with prior research in wearable assistive devices and haptic navigation systems [2], [4].

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